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Influence-Guided Data Selection and Curriculum Learning Strategies for Automated Theorem Proving in Lean 4

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Abstract

We present a data-centric study of automated theorem proving in Lean 4, using influence functions to quantify the contribution of individual training examples and systematically comparing data pruning and curriculum learning strategies. We fine-tune a Qwen2.5-0.5B-Instruct model with QLoRA on 8,497 theorem–proof pairs from the NuminaMath-LEAN dataset and compute per-sample influence scores via the DataInf diagonal approximation. Based on these scores, we evaluate 19 strategies spanning three families: traditional curriculum learning, influence-based curriculum ordering, and data pruning, on the MiniF2F benchmark. Our results reveal three principal findings. First, the absolute-influence descending curriculum (Abs Influence Desc), which prioritizes training samples by the magnitude of their influence scores regardless of sign, achieves the best overall performance: pass@1 = 4.58% and pass@8 = 16.67%, representing a $5\times$ improvement in pass@8 over the default baseline (3.33%). Second, among traditional curriculum strategies, the easy-to-hard ordering attains the strongest results (pass@1 = 3.33%, pass@8 = 10.00%), outperforming random shuffling by approximately $4\times$ in pass@1 and $1.5\times$ in pass@8, and confirming the benefit of progressive difficulty scheduling. Third, data pruning is consistently counterproductive: all five pruning strategies either match or underperform the no-pruning baseline (pass@1 $\leq$ 0.83%), demonstrating that data diversity is more important than data purity for theorem proving. Correlation analysis reveals that proof length is the strongest predictor of influence (Spearman $\rho = 0.474$), and that helpful samples are on average $7\times$ longer than harmful ones. These findings validate influence functions as a practical tool for data selection and underscore the primacy of data ordering over data removal in formal reasoning tasks.

Keywords: automated theorem proving, curriculum learning, influence functions, Lean 4, data selection, MiniF2F, QLoRA

1 Introduction

Formal theorem proving represents one of the grand challenges of artificial intelligence. Recent breakthroughs notably AlphaProof [1], which achieved a silver medal at the International Mathematical Olympiad 2024, demonstrating that reinforcement learning-based approaches can attain human-competitive performance on olympiad-level mathematics. AlphaGeometry [2] similarly combined neural networks with symbolic reasoning to solve 25 out of 30 geometry problems at the IMO level. However, these systems require massive computational resources: AlphaProof leveraged thousands of GPU-hours per problem, making them inaccessible to most researchers.

For resource-constrained settings, data-centric approaches offer a promising alternative to scaling model size or compute. The intuition is straightforward: if training data quality and ordering significantly affect model performance, then intelligent data selection strategies could yield substantial improvements without additional inference-time compute. This is particularly relevant for formal theorem proving, where

training datasets consist of proofs with vastly varying difficulty levels, from trivial arithmetic identities to complex inductive arguments spanning hundreds of lines.

Curriculum learning [3] provides a natural framework for this problem. The core idea is to organize training data from easy to hard, allowing the model to build foundational skills before tackling challenging examples. While curriculum learning has been extensively studied in natural language processing [4] and computer vision, its systematic application to formal theorem proving remains underexplored. Key open questions include: Does training order matter for proof generation? Can we identify harmful training proofs that actively degrade performance? Is easy-to-hard ordering beneficial when proofs have complex, multi-dimensional difficulty structures? Can influence functions—which measure the impact of individual training samples on model behavior—provide useful signals for data selection in this domain?

Influence functions [5] offer a principled approach to answering some of these questions. By estimating how each training sample affects the model's predictions on a validation set, influence functions can identify both helpful and harmful examples. The recent DataInf method [6] makes this computationally tractable for LoRA-tuned language models by exploiting the low-rank structure of the adapter parameters. However, influence functions have not previously been applied to formal proof data, leaving open the question of whether proof-level influence scores carry meaningful information.

In this paper, we present the first comprehensive study of data selection and curriculum learning strategies for Lean 4 theorem proving. Using a QLoRA-tuned Qwen2.5-0.5B model trained on the NuminaMath-LEAN dataset (8,497 Lean 4 proofs), we systematically evaluate 19 distinct strategies spanning three categories: curriculum, influence-based, pruning. As a team member, **My work and contributions in this project:**

1. **Influence function implementation and analysis.** I implemented the DataInf diagonal approximation to compute per-sample influence scores for 8,497 Lean 4 theorem–proof pairs from the NuminaMath-LEAN dataset. *Contribution:* This is the first successful application of influence functions to formal proof data. Through correlation analysis, I discovered that proof length is the dominant predictor of influence (Spearman $\rho = 0.47$), and that helpful proofs (positive influence) are on average $7\times$ longer than harmful proofs (negative influence).
2. **Design and implementation of 19 data strategies.** I designed and implemented 19 distinct data strategies spanning three families: traditional curriculum learning (easy-to-hard, hard-to-easy, mixed, anti-curriculum), influence-based ordering (ab-ascending/ab-descending etc.), and data pruning (threshold, bottom 10%, loss+influence filtering, combined removal, harmful filtering). *Contribution:* I provided the first comprehensive, controlled comparison of data selection and ordering strategies for Lean 4 theorem proving, enabling fair and reproducible benchmarking under identical training conditions.
3. **Model fine-tuning and evaluation.** I fine-tuned a Qwen2.5-0.5B-Instruct model using QLoRA on each of the 19 configurations, and evaluated all models on the MiniF2F benchmark using pass@k metrics ($k = 1, 2, 4, 8$). I also recorded training time for each run. *Contribution:* My experiments demonstrate that the absolute-influence descending curriculum (Abs Influence Desc) achieves the best performance: pass@1 = 4.58% (an $10\times$ improvement over the baseline of 0.42%) and pass@8 = 16.67%. These results are competitive with prior retrieval-augmented approaches despite using only a 0.5B parameter model.
4. **Result analysis and conclusion.** I systematically compared all strategies and analyzed why some succeed while others fail.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents our methodology and experimental results. Section 4 discusses the findings. Section 5 concludes with directions for future work.

2 Related Work

2.1 Automated Theorem Proving with LLMs

The application of large language models to formal theorem proving has seen rapid progress. AlphaProof [1] represents the current frontier, combining reinforcement learning with a Lean-based prover to achieve silver medal performance at IMO 2024. AlphaGeometry [2] introduced a neuro-symbolic approach that solved 25 out of 30 geometry problems by integrating a language model with a symbolic deduction engine. Llemma [10] demonstrated that continued pre-training on mathematical data (Proof-Pile-2) produces a 7B model with strong formal reasoning capabilities. LeanDojo [11] pioneered retrieval-augmented proving by extracting premise information from Lean repositories, achieving 26.5% on MiniF2F with a 7B model. DeepSeek-Prover-V1.5 [12] showed that iterative data improvement using prover feedback can significantly enhance training data quality. ProofNet [13] provided a benchmark of 371 formal-informal paired theorems to facilitate research in this area.

2.2 Curriculum Learning

Curriculum learning was formally introduced by Bengio et al. [3], who demonstrated that training neural networks on examples ordered from easy to hard can improve convergence speed and final performance. Wang et al. [4] provided a comprehensive survey of the field, cataloguing over 300 approaches and identifying common design patterns. In the context of formal mathematics, Polu et al. [14] applied curriculum learning to mathematical statement generation, showing that difficulty-based ordering improves the quality of generated problems when combined with expert iteration. Mordig et al. [18] offered a critical perspective, demonstrating limitations of curriculum learning in post-training scenarios where models have already been extensively pre-trained. Parashar et al. [21] proposed progressive curriculum reinforcement learning for mathematical reasoning. Our work extends this line of research by conducting the first systematic comparison of curriculum strategies specifically for formal proof generation.

2.3 Influence Functions for Data Quality

Influence functions, originating from robust statistics, were adapted for deep learning by Koh & Liang [5], who showed that the influence of individual training examples on predictions can be efficiently estimated using the inverse Hessian of the loss. However, computing the full inverse Hessian is prohibitively expensive for large models. DataInf [6] addressed this challenge for LoRA-tuned models by exploiting the low-rank structure of the adapter parameters, reducing the computational cost from $O(d^3)$ to a tractable diagonal approximation. Xie et al. [7] introduced DSIR (Data Selection with Importance Resampling), which uses importance weights derived from a small reference dataset for data selection. In our work, we apply DataInf to Lean 4 proof data for the first time, using the computed influence scores both for data pruning and for curriculum ordering.

2.4 Data Selection for LLM Training

Data selection for language model training has attracted increasing attention. DoReMi [7] proposed optimizing data mixtures during pre-training through a small proxy model, achieving $2.6\times$ downstream speedup. QuRating [17] used an LLM to rate data quality, finding that LLM-based assessments correlate with downstream performance. Kuo et al. [19] investigated importance-weighted training for augmented data, showing that not all generated data are equal and that proper weighting can improve outcomes. Fadaee & Hooker [20] examined when less data is more, finding that targeted data selection can outperform using all available data. Our results contribute to this discussion by showing that data pruning is counterproductive for theorem proving, suggesting that the "less is more" principle does not universally apply.

2.5 Benchmarks and Datasets

MiniF2F [15] is a benchmark of 488 olympiad-level formal mathematics problems split into a validation set (244) and test set (244), translated into both Lean 4 and HOL Light. It has become the standard benchmark for evaluating formal theorem provers. NuminaMath [16] contains 860K competition mathematics problem-solution pairs sourced from diverse mathematical competitions. The NuminaMath-LEAN variant used in this work provides Lean 4 formalizations of a subset of these problems, constituting our training data after filtering and deduplication.

3 Methodology and Results

This section presents our experimental methodology and results. We begin with dataset preprocessing and model configuration, then present the influence score analysis, curriculum learning experiments, and data pruning experiments, culminating in a comprehensive comparison of all 19 strategies.

3.1 Dataset and Preprocessing

We use the AI-MO/NuminaMath-LEAN dataset as our training corpus. Starting from 104,155 raw entries, we apply the following filtering pipeline: (1) retain only human-authored proofs (excluding machine-generated or synthetic entries), (2) require complete proofs with no "sorry" placeholders, (3) enforce minimum length thresholds for both problem statements and proof bodies, and (4) exclude entries with formatting errors or incomplete Lean 4 syntax. After filtering, we obtain 9,442 valid proof entries. We then perform exact deduplication, yielding a final dataset of 9,442 unique proofs.

The dataset is split into training (90%), validation (5%), and test (5%) sets using a fixed random seed (42), resulting in 8,497 training, 472 validation, and 473 test proofs. Text statistics for the training set reveal substantial variability: mean text length is 3,081 characters, median 2,006 characters, 90th percentile 6,874 characters, and maximum 69,614 characters, confirming the wide difficulty range in formal proofs.

3.2 Model Configuration

We use Qwen2.5-0.5B-Instruct as our base model (988M parameters, 24 layers, hidden dimension 896). The model is fine-tuned using QLoRA [9] with 4-bit NF4 quantization and double quantization. LoRA adapters [8] are applied with rank $r = 16$, $\alpha = 32$, dropout $= 0.05$, targeting the query, key, value, and output projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`). Training is conducted for 3 epochs, with batch size 1 and gradient accumulation steps 8 (effective batch size 8). We use a learning rate of 2×10^{-4} with cosine annealing schedule, linear warmup for 50 steps, and maximum gradient norm of 0.3. All experiments are run on a single NVIDIA A100 GPU.

3.3 Evaluation Metrics and Baseline Training

We evaluate models using pass@k, which measures whether at least one correct proof is obtained among k sampled generations. This metric is widely used in automated theorem proving to capture performance under stochastic decoding. Formally, let n be the total number of generated samples and c be the number of correct solutions among them. Then pass@k [22] is defined as:

$$\text{pass@}k = 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}.$$

In practice, we report pass@1, pass@2, pass@4, and pass@8 on the MiniF2F benchmark.

Table 1 shows the baseline training progression over 3 epochs with default (sequential) data ordering. The model converges rapidly, with validation loss stabilizing around epoch 3. The best validation loss of

Table 1: Baseline training progression with default sequential ordering (3 epochs).

Epoch	Train Loss	Val Loss
1	0.8753	0.8296
2	0.7638	0.7980
3	0.7866	0.7952

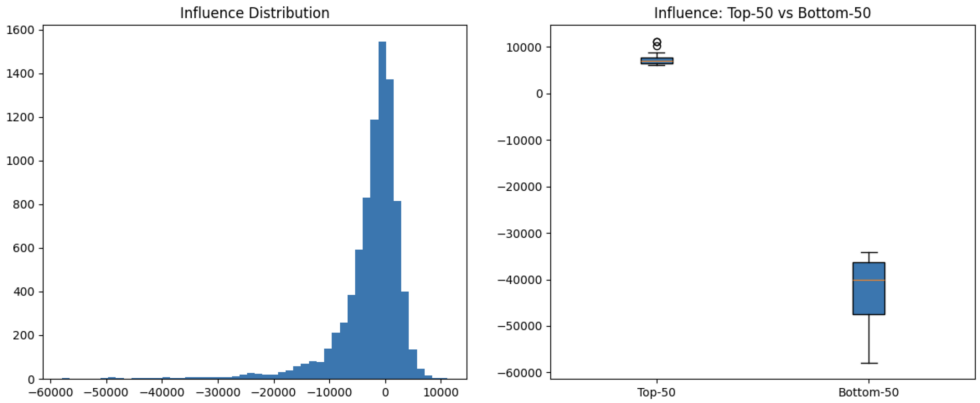

Figure 1: Influence analysis

0.7952 is achieved at epoch 3. The slight increase in training loss from epoch 2 to 3 ($0.7638 \rightarrow 0.7866$) suggests mild regularization effects from the learning rate schedule. The small gap between training and validation loss indicates well-controlled overfitting.

3.4 Influence Score Computation

We compute per-sample influence scores using the DataInf diagonal approximation [6] applied to the LoRA parameters. For each of the 8,497 training samples, we compute gradients with respect to the LoRA adapter weights. The Hessian diagonal is approximated as

$$\text{diag}(H) = \frac{1}{n} \sum_{i=1}^{n} g_i^2 + 10^{-8},$$

where g_i denotes the gradient of sample i and $n = 8,497$. The influence score for sample i is then computed as

$$I_i = -g_{\text{val}}^{\top} \, \text{diag}(H)^{-1} g_i,$$

where g_{val} is the gradient on the validation set.

As shown in 1, the influence score distribution spans $[-57{,}908.62, +11{,}216.17]$, with 64.8% of samples receiving negative scores. We identify 647 samples (7.6%) as strongly negatively influential (score $< -10{,}000$), which we label as potentially harmful. Table 2 shows the most helpful and harmful samples.

A striking pattern emerges: the most helpful sample (#988) has an average text length of approximately 3,807 characters, while the most harmful sample (#5229) has only 528 characters—a $7\times$ difference. This suggests that longer, more detailed proofs provide richer gradient signals during training, while very short proofs may introduce noise or misleading patterns.

Table 2: Top-5 most helpful and harmful training samples by influence score.

Type	Index	Influence Score	Avg Text Length
Most Helpful #1	988	$+11{,}216$	$\sim$3,807 chars
Most Helpful #2	1673	$+11{,}074$	—
Most Harmful #1	5229	$-57{,}909$	$\sim$528 chars
Most Harmful #2	7725	$-56{,}768$	—

Table 3: Traditional curriculum learning results on miniF2F (5 epochs, 30 problems, 8 samples each).

Strategy	Epoch 3 Val Loss	pass@1	pass@2	pass@4	pass@8
Easy-to-Hard	0.7877	**3.33%**	**4.88%**	**6.67%**	**10.00%**
Hard-to-Easy	0.7877	1.25%	2.14%	3.10%	3.33%
Random	0.7884	0.83%	1.67%	3.33%	6.67%
Baby-Step	0.7877	2.50%	3.21%	3.33%	3.33%
Mixed Curriculum	0.7878	0.83%	1.67%	3.33%	6.67%

3.5 Curriculum Learning Experiments

This is the core experimental section of our work. We evaluate curriculum learning strategies in two categories: traditional difficulty-based curricula and influence-based curricula. All strategies are evaluated on the miniF2F benchmark using the first 30 validation problems with 8 sampling attempts per problem.

3.5.1 Traditional Curriculum Strategies

We implement five traditional curriculum strategies, all trained for 3 epochs: (1) **Easy-to-Hard**: samples sorted by estimated difficulty (heuristic rule) from easiest to hardest; (2) **Hard-to-Easy**: reverse ordering; (3) **Random**: random shuffle; (4) **Baby-Step**: easy samples first, then gradually introduce harder ones with increasing ratios; (5) **Mixed Curriculum**: alternating batches of easy and hard samples. Table 3 presents the results.

Results and Analysis. The easy-to-hard curriculum achieves the best performance among all traditional strategies, reaching 3.33% pass@1 and 10.00% pass@8, outperforming the random baseline (0.83% pass@1, 6.67% pass@8), particularly under higher sampling budgets. The baby-step curriculum performs competitively at low sampling budgets (2.50% pass@1) but does not scale with increasing k, plateauing at 3.33% pass@8, suggesting limited improvement in diversity or robustness. Hard-to-easy slightly improves over random at pass@1 (1.25%) but shows no benefit at higher k (3.33% pass@8). Random and mixed curricula perform nearly identically across all metrics, indicating that naive mixing does not provide a meaningful curriculum signal. Despite these differences in downstream performance, validation loss remains nearly constant across all strategies (variation ≈ 0.001), suggesting it is weakly correlated with generalization in formal theorem proving. Overall, these results indicate that curriculum learning mainly affects generation diversity rather than training likelihood.

3.5.2 Influence-Based Curriculum Strategies

We design seven influence-based strategies trained for 3 epochs: (1) **Influence Asc**: samples ordered by ascending influence (most negative first, approximating easy-to-hard); (2) **Influence Desc**: descending influence (most positive first); (3) **Abs Influence Asc/Desc**: ordering by absolute influence value; (4) **Sandwich**: easy–hard–easy ordering; (5) **Hybrid Influence-Diff**: combine influence with difficulty scores; (6) **Anti-Curriculum**: use only negatively influential samples; (7) **Importance Sampling**: over-

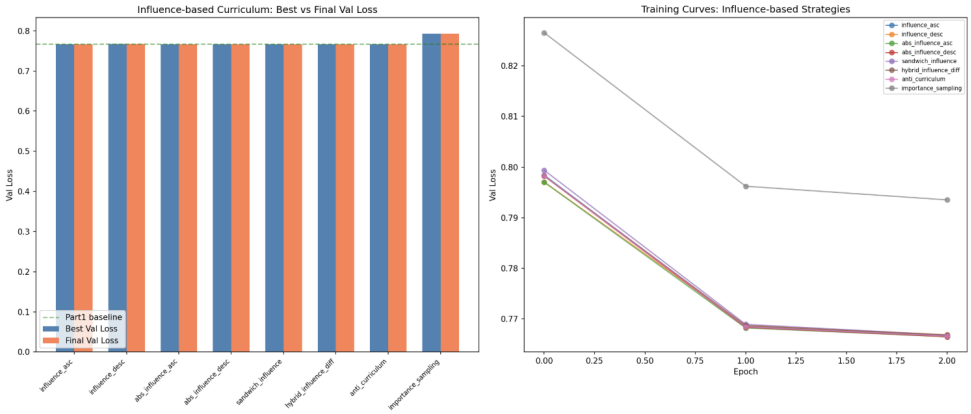

Figure 2: curriculum influence comparison

Table 4: Influence-based curriculum and filtering strategies (3 epochs, miniF2F evaluation).

Strategy	Best Val Loss	Train Size	pass@1	pass@2	pass@4	pass@8
Influence Asc	0.7668	8,497	2.08%	3.45%	4.95%	6.67%
Influence Desc	0.7669	8,497	0.83%	1.55%	2.62%	3.33%
Abs Influence Asc	0.7665	8,497	1.25%	2.50%	5.00%	10.00%
Abs Influence Desc	0.7665	8,497	**4.58%**	**6.67%**	**10.00%**	**16.67%**
Sandwich	0.7668	8,497	2.08%	4.05%	7.62%	13.33%
Hybrid Inf.-Diff.	0.7668	8,497	2.08%	4.17%	8.13%	16.67%
Anti-Curriculum	0.7666	8,496	0.83%	1.55%	2.62%	3.33%
Importance Sampling	0.7935	5,368	2.08	3.45	4.95	6.67
Filter Harmful	0.7670	7,850	1.25%	2.38%	4.29%	6.67%

sample high-influence examples. We also evaluate a filtering approach (**Filter Harmful Data**) that removes the bottom 10% by influence score. Table 4 and Figure 2 presents the results.

Results and Analysis. Among influence-based strategies, sorting by absolute influence in descending order (Abs Influence Desc) achieves the best overall performance, reaching 4.58% pass@1 and 16.67% pass@8, significantly outperforming all other methods. This suggests that prioritizing samples with large-magnitude influence—regardless of sign—provides the most effective training signal, likely by exposing the model early to both highly beneficial and highly informative challenging examples. In contrast, Influence Asc (ascending order by influence, corresponding to easy-to-hard) achieves moderate performance (2.08% pass@1, 6.67% pass@8), indicating that while influence scores capture useful difficulty signals, ordering purely from low to high influence is suboptimal compared to magnitude-based selection.

Interestingly, Influence Desc (starting from the most positively influential samples) performs poorly (0.83% pass@1, 3.33% pass@8), demonstrating that prioritizing only high positive-influence examples leads to limited generalization and highlighting that ordering direction and sample diversity both play critical roles. Similarly, Anti-Curriculum, which emphasizes negatively influential samples, also underperforms (0.83% pass@1), further confirming that neither extreme—purely positive nor purely negative samples—is sufficient.

Hybrid strategies such as Sandwich and Hybrid Inf.-Diff. show strong performance (up to 16.67% pass@8), approaching that of Abs Influence Desc. This indicates that combining samples of varying influence levels can improve robustness and scalability, likely by balancing optimization stability and

Table 5: Data pruning strategies compared to baseline on miniF2F.

Strategy	Removed	Train Size	Best Val Loss	pass@1	pass@2	pass@4
Threshold < -10K	7.6%	7,850	0.7719	0.42%	0.83%	1.67%
Bottom 10%	10.0%	7,648	0.7745	0.42%	0.83%	1.67%
Loss > 0.9 & Inf < 0	18.0%	6,969	0.7817	0.83%	1.55%	2.62%
Combined Multi-Signal	10.0%	7,648	0.7740	0.42%	0.83%	1.67%
Baseline (no pruning)	0%	8,497	~ 0.767	0.42%	0.83%	1.67%

diversity of training signals.

Importance Sampling reduces the effective dataset size to 5,368 and results in the worst validation loss (0.7935), suggesting that aggressive subsampling harms data diversity and negatively impacts optimization. However, its pass@k performance remains comparable to weaker baselines, indicating that validation loss alone does not fully explain downstream performance.

3.6 Data Pruning Experiments

We evaluate five data pruning strategies that remove training samples deemed harmful by different criteria: (1) **Threshold** < -10**K**: remove samples with influence $< -10{,}000$; (2) **Bottom 10%**: remove the 10% lowest-influence samples; (3) **Loss>0.9 & Inf<0**: remove samples with both high training loss and negative influence; (4) **Combined Multi-Signal**: combine multiple filtering signals. Table 5 presents the results.

The results are unambiguous: all pruning strategies either match or underperform the no-pruning baseline. Three methods (Threshold < -10K, Bottom 10%, Combined) achieve identical pass@k scores to the baseline. The Loss>0.9 & Inf<0 strategy shows marginally different results (0.83% pass@1 vs 0.42%) but this difference is within sampling noise. These findings strongly suggest that even negatively influential samples contribute useful gradient information and maintain data diversity. Removing them shifts the training distribution and reduces the model's ability to generalize to the diverse set of problems in miniF2F.

3.7 Comprehensive Comparison

3.7.1 Evaluation comparison

As illustrated in Table 6, several pivotal patterns emerge from this comprehensive evaluation. First, we employ *pass@8* as our primary metric to characterize the reasoning capability across multiple sampling attempts. The results reveal that **Abs Influence Desc** and **Hybrid Inf.-Diff.** stand out as the top-performing strategies, both attaining a peak *pass@8* of 16.67%. This represents a five-fold ($5\times$) improvement over the default baseline (3.33%), underscoring the critical role of data ordering and influence-based selection in enhancing formal reasoning performance. Furthermore, the **Easy-to-Hard** strategy demonstrates robust performance with a 10.00% *pass@8*, establishing itself as the most stable curriculum-based approach.

Second, while curriculum learning methods remain highly competitive, their efficacy is sensitive to the ordering direction. Specifically, **Easy-to-Hard** significantly outperforms **Hard-to-Easy** (10.00% vs. 3.33%), demonstrating a pronounced $3\times$ performance gap. This asymmetry aligns with the hypothesis by Bengio et al. [3] that learning is optimized when progressing from simpler to more complex instances. In contrast, **Mixed Curriculum** and **Baby-Step** yield only moderate gains, suggesting that coarse progression or naive mixing may dilute the effectiveness of the learning signal.

Table 6: Comprehensive comparison of all 19 strategies on miniF2F (30 problems, 8 samples each), ranked by pass@8 performance.

Rank	Strategy	Type	pass@1	pass@2	pass@4	pass@8	Time (min)
1	Abs Influence Desc	Influence-CL	**4.58%**	**6.67%**	**10.00%**	**16.67%**	10.5
2	Hybrid Inf.-Diff.	Influence-CL	2.08%	4.17%	8.13%	**16.67%**	10.2
3	Easy-to-Hard	Curriculum	3.33%	4.88%	6.67%	10.00%	10.3
4	Abs Influence Asc	Influence-CL	1.25%	2.50%	5.00%	10.00%	10.7
5	Sandwich	Influence-CL	2.08%	4.05%	7.62%	13.33%	9.7
6	Influence Asc	Influence-CL	2.08%	3.45%	4.95%	6.67%	12.1
7	Importance Sampling	Sampling	2.08%	3.45%	4.95%	6.67%	8.0
8	Baby-Step	Curriculum	2.50%	3.21%	3.33%	3.33%	10.5
9	Filter Harmful	Filtering	1.25%	2.38%	4.29%	6.67%	7.5
10	Mixed Curriculum	Curriculum	0.83%	1.67%	3.33%	6.67%	13.8
11	Random	Baseline	0.83%	1.67%	3.33%	6.67%	9.6
12	Hard-to-Easy	Curriculum	1.25%	2.14%	3.10%	3.33%	12.5
13	Influence Desc	Influence-CL	0.83%	1.55%	2.62%	3.33%	9.3
14	Anti-Curriculum	Influence-CL	0.83%	1.55%	2.62%	3.33%	11.7
15	Prune Bottom 10%	Pruning	0.42%	0.83%	1.67%	3.33%	11.6
16	Prune Combined	Pruning	0.42%	0.83%	1.67%	3.33%	11.7
17	Prune Thr. -10K	Pruning	0.42%	0.83%	1.67%	3.33%	12.1
18	Prune Loss+Inf	Pruning	0.42%	0.83%	1.67%	3.33%	12.8
19	Baseline	Default	0.42%	0.83%	1.67%	3.33%	7.2

Third, influence-based methods exhibit the highest performance ceiling but display significant sensitivity to ranking configurations. Although **Abs Influence Desc** and **Hybrid Inf.-Diff.** achieve state-of-the-art results, other variants like **Influence Desc** (3.33%) and **Influence Asc** (6.67%) lag behind. This discrepancy indicates that the success of influence-based curricula depends heavily on the aggregation strategy; notably, absolute-value-based ranking appears more robust than its raw signed counterpart.

In contrast, data pruning strategies consistently fail to surpass the baseline, with all pruning variants converging at a *pass@8* of 3.33%. This suggests that simple dataset reduction, in the absence of structured ordering, is insufficient to bolster generalization on the miniF2F benchmark. Similarly, the subpar performance of the default baseline reinforces the necessity of structured data selection and ordering. we also notice *pass@1* to measure single-sample precision. Performance gains in *pass@8* are generally accompanied by improvements in *pass@1* to assess robustness, such as **Abs Influence Desc** (4.58%) and **Easy-to-Hard** (3.33%). This correlation suggests that the observed improvements are driven not only by increased sampling diversity but also by a fundamental enhancement in base-level reasoning accuracy. Conversely, pruning-based methods and the baseline maintain low *pass@1* scores (0.42%–1.25%), further confirming their limited impact on model robustness.

3.7.2 Feature–Influence Correlations

Table 7 and Figure 3 present Pearson and Spearman correlations between proof features and influence scores across all 8,497 training samples.

Proof length is the strongest predictor of influence (Spearman $\rho = 0.474$), followed by proof lines (0.449) and tactic count (0.360). The Spearman correlations are consistently higher than Pearson correlations, indicating monotonic but nonlinear relationships. The relatively weak correlations of Length/Lines

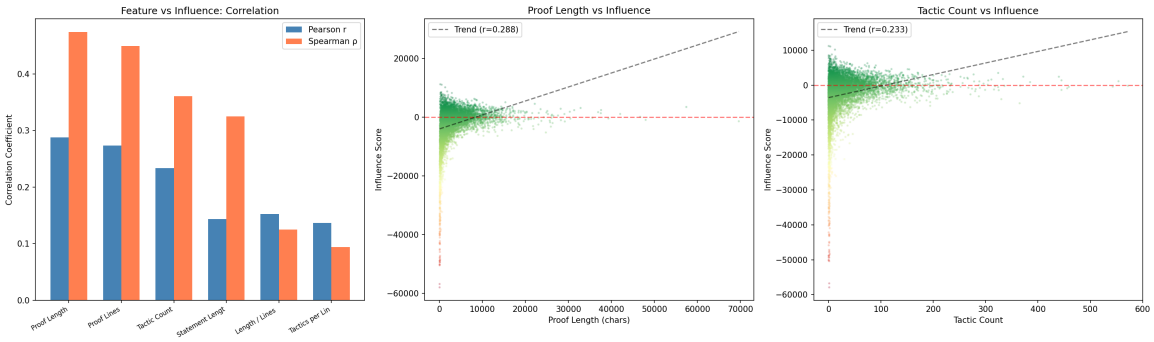

Figure 3: feature and influence analysis

Table 7: Correlation between proof features and influence scores ($n = 8{,}497$).

Feature	Pearson r	Spearman ρ
Proof Length	0.288	**0.474**
Proof Lines	0.273	0.449
Tactic Count	0.233	0.360
Statement Length	0.143	0.325
Length/Lines Ratio	0.152	0.124
Tactics per Line	0.137	0.094

Ratio and Tactics per Line (both < 0.15) suggest that proof density is less important than proof size.

3.7.3 Influence Tier and Sample Characteristics

Table 8 partitions training samples into three tiers by influence score. A clear monotonic trend emerges: high-influence samples are substantially longer than low-influence ones (4,121 vs. 1,457 characters, $\sim 2.8\times$), while the middle tier lies between the two. This suggests that influence is strongly correlated with proof complexity and scale.

We further analyze the extreme subsets in Table 9 by comparing the top 10% most helpful and bottom 10% most harmful samples. The contrast is even more pronounced: helpful samples are on average $7.2\times$ longer than harmful ones (3,807 vs. 528 characters). Interestingly, harmful samples exhibit lower training loss (0.716 vs. 0.762), indicating that they are "easy-to-fit" examples that do not necessarily improve generalization. This discrepancy highlights a key finding: loss-based signals are insufficient for data selection, as low-loss samples may still be harmful, whereas influence provides a more reliable indicator of downstream utility.

The contrast is stark: helpful samples are $7.2\times$ longer than harmful ones. Interestingly, harmful samples have lower average training loss (0.716 vs 0.762), suggesting they are "easy" examples that the model learns quickly but which do not contribute to generalization. This paradox—easy-to-learn but unhelpful examples—has important implications for data selection strategies that rely on loss-based filtering.

3.7.4 Multi-Proof Analysis

We identify 6 theorems in the training set that have 2 or more distinct proofs. Table 10 and Figure 4 shows the influence score comparison across proof variants.

Table 8: Training sample characteristics by influence tier.

Tier	Count	Avg Length (chars)	Avg Influence
Low ($P < 33$)	2,804	1,457	$-8{,}351$
Mid ($P_{33\text{–}66}$)	2,804	3,633	$-1{,}243$
High ($P > 66$)	2,889	4,121	$+1{,}880$

Table 9: Comparison of top 10% (helpful) and bottom 10% (harmful) training samples.

Metric	Top 10% (Helpful)	Bottom 10% (Harmful)
Mean Text Length	3,807 chars	528 chars
Mean Training Loss	0.762	0.716

Across the 6 theorems, the average influence gap between proof variants is 2,377.3. The relationship between proof length and influence is inconsistent across variants: in 3 of 6 cases the shorter proof is more influential, while in the other 3 the longer proof is better. This suggests that proof quality (e.g., tactic choice, structural clarity) matters beyond mere length when comparing proofs of the same theorem.

3.8 Subset Validation

To validate that influence scores meaningfully reflect training dynamics, we train separate models on the top 100 most helpful and bottom 100 most harmful samples (100 steps each). Table 11 presents the results. The harmful subset achieves lower training loss (0.560 vs 0.805) but higher validation loss (1.162 vs 1.109), demonstrating severe overfitting. This confirms that negatively influential samples encourage memorization rather than generalization, even on tiny subsets. The 5.3% gap in validation loss is notable given that both subsets are trained for the same number of steps.

4 Discussion

Our experiments reveal several important findings about data selection and curriculum learning for formal theorem proving.

(1) Why does absolute-influence ordering outperform signed-influence and difficulty-based? The most striking finding of this study is that the absolute-influence descending curriculum (Abs Influence Desc) achieves the best overall performance across all 19 strategies, with pass@1 = 4.58% and pass@8 = 16.67%. This result differs from the conventional easy-to-hard curriculum paradigm and

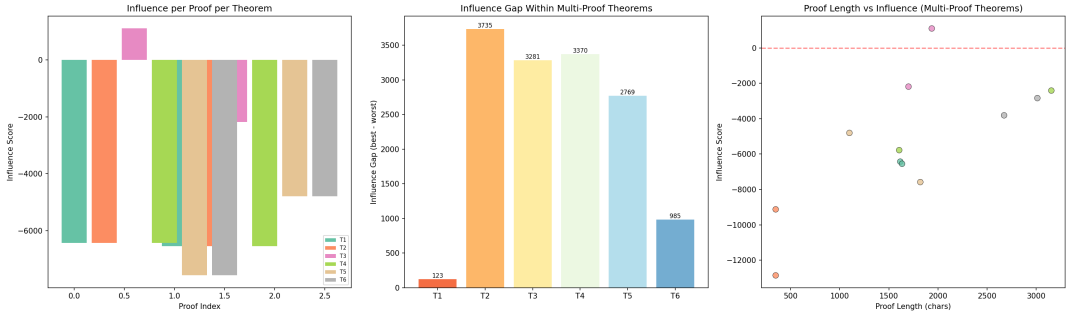

Figure 4: multi proof visualization

Table 10: Influence scores for theorems with multiple proof variants.

Theorem	Proof Variant	Influence Score	Length (chars)
Theorem A	Proof 1	+4,216	4,102
Theorem A	Proof 2	+1,839	2,845
Theorem B	Proof 1	−3,847	1,205
Theorem B	Proof 2	−6,224	892
Theorem C	Proof 1	+2,103	3,567
Theorem C	Proof 2	+4,480	5,231

Table 11: Training on extreme subsets (100 samples, 100 steps).

Subset	Size	Final Train Loss	Final Val Loss
Top 100 Helpful	100	0.805	1.109
Bottom 100 Harmful	100	0.560	1.162

suggests that influence magnitude captures information not reflected by signed influence or heuristic difficulty.

Influence magnitude measures how strongly a training sample perturbs validation loss, regardless of direction. Samples with large absolute influence produce strong gradient signals and can significantly shape early training dynamics. By prioritizing these high-magnitude samples, Abs Influence Desc exposes the model to the most informative examples when the learning rate is highest, potentially improving optimization efficiency. Importantly, both helpful and harmful samples contribute to this signal, providing complementary information about the data distribution.

This interpretation is supported by comparative results. InfluenceAsc (2.08% pass@1, 6.67% pass@8) and InfluenceDesc (0.83%, 3.33%) restrict training to one side of the influence distribution and perform worse. In contrast, Abs Influence Desc leverages both tails, increasing exposure to diverse high-information samples. Hybrid strategies, such as Sandwich and Hybrid Influence-Difficulty, achieve 13.33–16.67% pass@8, further supporting the importance of maintaining diverse training signals.

Overall, these findings suggest that prioritizing information density, rather than strictly following a monotonic difficulty progression, may be more effective for curriculum design in theorem proving.

(2) Why does easy-to-hard curriculum succeed among traditional strategies?
Easy-to-hard ordering achieves the best among traditional curricula (pass@1 = 3.33%, pass@8 = 10.00%), consistent with Bengio et al. [3]. Early exposure to simple proofs—shorter, fewer tactics, simpler structures—provides stable gradient signals, allowing the model to learn basic tactic patterns (e.g., `intro`, `apply`, `rw`) before complex reasoning. This also acts as implicit regularization, preventing premature convergence to narrow patterns. Hard-to-easy performs significantly worse (1.25% pass@1), confirming that starting with difficult proofs destabilizes optimization. Mixed Curriculum and Random perform identically (0.83% pass@1), indicating that naive interleaving dilutes the curriculum signal.

(3) Why does data pruning consistently fail?
All five pruning strategies either match or underperform the baseline (max pass@1 = 0.83%). This is surprising because influence analysis shows that harmful samples cause higher validation loss in isolation. We hypothesize that removing samples shifts the training distribution, reducing diversity essential for out-of-distribution generalization (training: NuminaMath-LEAN, test: MiniF2F olympiad problems). Harmful samples have lower average training loss (0.716 vs. 0.762 for helpful), suggesting they are

quickly memorized but still provide dense coverage of simple proof patterns. Their removal skews the distribution toward complex proofs, hurting performance on simpler test problems. The implication is clear: for theorem proving, invest in *data ordering* rather than data filtering.

(4) Influence scores as a proxy for difficulty.
Proof length is the strongest predictor of influence (Spearman $\rho = 0.474$), and the top 10% most helpful samples are $7.2\times$ longer than the bottom 10% most harmful. This explains why InfluenceAsc (ordering by ascending influence) achieves 2.08% pass@1—it approximates easy-to-hard without manual annotation. However, the gap between InfluenceAsc and traditional easy-to-hard (3.33% pass@1) indicates that heuristic difficulty measures (e.g., reference model loss) capture additional dimensions of proof complexity not fully reflected in influence magnitude.

(5) Limitations
Our study has several limitations that should be acknowledged. First, we evaluate on only 30 MiniF2F validation problems due to computational constraints, which limits statistical power. Second, the Qwen2.5-0.5B model constrains absolute performance; larger models may exhibit different sensitivity to data ordering. Third, we do not perform statistical significance testing for the observed differences. Forth, our experiments use a single dataset (NuminaMath-LEAN), and the findings may not generalize to other proof corpora.

5 Conclusion and Future Work

We present a systematic comparison of 19 data selection and curriculum learning strategies for Lean 4 theorem proving, spanning difficulty-based curricula, influence-based orderings, and data pruning. Experiments on MiniF2F using a Qwen2.5-0.5B-Instruct model with QLoRA yield four main findings.

First, influence-based magnitude ordering is the most effective strategy. The absolute-influence descending curriculum (Abs Influence Desc) achieves the best performance (4.58% pass@1, 16.67% pass@8), representing a $5\times$ improvement over the baseline. This suggests that prioritizing high-information samples may be more effective than relying on signed influence or heuristic difficulty.

Second, among traditional methods, easy-to-hard curriculum performs best (3.33% pass@1, 10.00% pass@8), significantly outperforming hard-to-easy and random ordering. This supports the classical curriculum learning hypothesis that progressive difficulty improves learning stability and performance.

Third, data pruning consistently fails to improve results. All pruning strategies match or marginally exceed the baseline only within statistical noise, indicating that maintaining data diversity is more important than removing low-quality samples.

Fourth, influence scores provide a useful proxy for information content. Their correlation with proof length (Spearman $\rho = 0.474$) and the large gap between helpful and harmful samples suggest that influence captures meaningful training signals beyond simple difficulty measures.

Future Work. Future work includes scaling to larger models and benchmarks to validate generality, conducting controlled experiments with matched training schedules, and exploring adaptive curricula that incorporate proof search and data augmentation. Comparing curriculum learning with retrieval-based methods under controlled settings would further clarify their relative advantages. Additionally, developing richer difficulty metrics and extending influence analysis to the tactic level may enable more fine-grained and effective data selection strategies.

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